

DIANA MAYALO

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Data Scientist & AI Automation Engineer

Actuarial Science graduate and Data Scientist & AI Automation Engineer with hands-on experience building machine learning systems, AI applications, automated data pipelines, predictive models, and production-ready APIs. Skilled in taking solutions from data collection and feature engineering through model development, backend integration, automation, and deployment using Python, SQL, FastAPI, Django REST, Docker, XGBoost, and TensorFlow. Experienced in transforming complex data into practical, scalable products and business insights.

EDUCATION

JKUAT | Bachelor of Science in Actuarial Science

Sep 2019-June 2024 (Graduated June 2024)

Moringa School | Data Science Bootcamp

Feb 2025 - Aug 2025 (Graduated Nov 2025)

TECHNICAL SKILLS

- **Machine Learning & AI**
- Supervised Learning | Classification | Regression | Predictive Modeling | Gradient Boosting | XGBoost | Deep Learning | CNNs | NLP | Feature Engineering | Model Evaluation | Scikit-learn | TensorFlow | Keras
- **Data Engineering & Analytics**
- ETL Pipelines | Data Cleaning & Transformation | Data Quality | SQL | PostgreSQL | MySQL | Pandas | NumPy | Spark | Statistical Analysis | Exploratory Data Analysis | API Integration
- **AI Engineering & Backend**
- Python | FastAPI | Django REST Framework | REST APIs | LLM Integration | RAG | AI Agents | Conversation Systems | PostgreSQL | Backend Architecture | Async Python
- **Automation, MLOps & Deployment**
- Docker | Git | GitHub Actions | CI/CD | Linux/Bash | Model Deployment | Streamlit | Google Apps Script | Google Sheets API | Workflow Automation | AWS (EC2, S3, Lambda)
- **Visualization & Geospatial**
- Power BI | Plotly | Plotly Mapbox | Streamlit | Interactive Dashboards | Google Earth Engine | Geospatial Data Analysis | GIS.

TECHNICAL PROJECTS

CleanWater AI | ML Engineer & Data Pipeline Architect

Python | XGBoost | NLP | Pandas | Google Earth Engine | WPDx | GEMS API | ETL | Docker | Streamlit

- Engineered an end-to-end water-quality risk monitoring system integrating satellite imagery, crowdsourced water-point data, and external APIs across 22,000+ water points in Kenya.

- Built a multi-source ETL and data processing pipeline combining Google Earth Engine geospatial data with WPDx and GEMS datasets.
 - Developed and evaluated an XGBoost predictive model with NLP-based processing to identify potential water contamination risks.
 - Containerized the application with Docker and deployed an interactive Streamlit application for real-time monitoring and risk assessment.
 - Designed the system to support proactive identification of potentially unsafe water sources and data-driven public health interventions.
- [Live Demo](#)
- [Repo](#)

Support-AI | AI Engineer & Full-Stack Developer

Tech Stack: Python | FastAPI | React | PostgreSQL | LLMs | RAG | Docker

- Built a full-stack AI customer-support platform with LLM-powered chat, knowledge-base retrieval, conversation persistence, and human escalation.
 - Developed FastAPI REST APIs and PostgreSQL architecture for scalable conversation and customer-data management.
 - Implemented RAG and intelligent escalation for routine queries and complex support cases.
 - Containerized and deployed the application using Docker, demonstrating end-to-end AI product development.
- [Live Demo](#)

Nairobi Emergency Dashboard (NED) | AI & Backend Engineer

Python | Streamlit | Google Sheets API | Google Apps Script | Plotly | Mapbox | Data Automation

- Built and deployed a real-time emergency dashboard for Nairobi disaster-response workflows.
 - Automated field-report ingestion and processing using Google Sheets API, Google Apps Script, and Python.
 - Developed interactive Mapbox geospatial visualizations to identify incidents and prioritize response.
 - Supported real-time emergency coordination across Nairobi County.
- [Live Demo](#)

Crop Disease Classification | Machine Learning Engineer

Tech Stack: Python | TensorFlow | Keras | CNN | Computer Vision | Scikit-learn | Streamlit

- Developed a convolutional neural network (CNN) for multi-class classification of healthy and diseased crop leaves.
- Trained the model on the PlantVillage dataset containing 15,000+ crop images across 15 disease classes.

- Applied data augmentation and preprocessing techniques to improve model generalization and classification performance.
- Evaluated model performance using classification metrics and deployed the trained model through an interactive Streamlit application.
- Designed the solution to demonstrate how computer vision can support early agricultural disease detection.
→ [Repo](#)

EXPERIENCE

Data Analyst (Freelance), Intelinkpro, Remote

Oct 2023 – Apr 2025

- Automated Excel processing: Developed Python scripts to programmatically process complex client datasets, reducing manual report generation time by 30% while eliminating structural calculation errors
- Business Intelligence: Designed and deployed scalable Power BI dashboards for corporate clients, providing real-time operational metrics that improved decision-making efficiency and contributed to 15% improvement in target outcomes
- Data Preparation: Cleaned and structured messy transactional databases using SQL and Python, preparing high-quality datasets for BI and analytics pipelines
- Client Project Management: Managed end-to-end analytical workflows, ensuring timely delivery and alignment with business requirements across multiple concurrent projects

Data Analyst Intern, Future Interns, Bengaluru, India

Sep 2024 - Nov 2024

- Predictive Modeling: Built and validated high-performance loan eligibility classification model in Python, achieving 81.3% test accuracy through systematic feature engineering and model tuning
- Exploratory Data Analysis: Conducted deep statistical analysis on large datasets to identify patterns, distributions, and predictive features for model training
- Dashboard Development: Translated complex sports performance metrics into interactive Power BI dashboards, enabling stakeholders to extract tactical insights and make data-driven decisions
- Agile Environment: Executed iterative data sprints within agile workflows, collaborating with cross-functional teams to deliver rapid prototypes and insights

REFERENCES

Available upon request.